

The Impact of an Alternative Cloud Seeding Chemical, Aluminum Oxide, on the Growth of and Health of Photosystem 2 in a *Brassica Rapa* Model

Background



- Global **drought area has increased**
 - 78% of land is drier than 30 years ago
- Rising temperature
 - Ecosystems above optimal temperatures for photosynthesis**



https://images.newscientist.com/wp-content/uploads/2026/04/29171338/SEL_294983058.jpg

Cloud Seeding

- Technique used in drought areas to **enhance precipitation**
- Release chemicals in the atmosphere for water vapor to form around
- Glaciogenic - Typically uses Silver Iodide, Aluminum Oxide Alternative

Previous Research

- AgI doesn't impact organisms if seeding happens once
 - Minimal Accumulation**
 - No impact on photosynthesis
- Overwhelming evidence in favor of Al on plants**
 - Benefit of increasing plant growth
 - Mitigation of biotic and abiotic stress



https://en.wikipedia.org/wiki/Aluminum_oxide

Because aluminum oxide has a similar makeup as silver iodide, it has been proposed as an alternative seeding chemical. The mitigation of biotic and abiotic stress can help regulate photosynthetic levels, as photosynthesis is strongly impacted by stress.

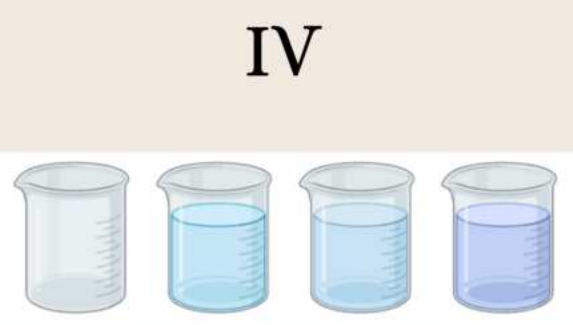
Purpose

Explore a possible alternative chemical to use for cloud seeding and its impacts, to lead to a **more environmentally conscious** cloud seeding practice.

Hypothesis and Variables

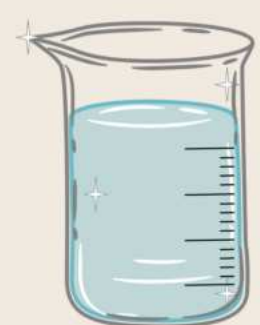
Hypothesis

Aluminum oxide will **increase** plant growth and photosynthetic activity because very small amounts of aluminum oxide can stimulate root development and enhance nutrient uptake.



Concentration of Aluminum Oxide: 0.005 & 0.08 micrograms/L

Control



Water without added chemicals

DV



Height (cm), # of leaves, Quench value, F value

Methodology

Setting up Hydroponic System

1 Place Air Stones in Bin



Place air stones with attached tubes inside the clear plastic bin. Connect the tubes to an air pump outside the bin.

2 Add Rock Wool to Net Pots



Dampen the rock wool cubes. Place one cube into each net pot in the lid.

3 Mix Nutrient Solution



Fill the bin with 8L of water. Add 8mL of plant nutrient solution to the water.

4 Place Lid & Start System



Place the lid with net pots on the bin. Keep the system under constant light and air from the pump.

Graphic created with researcher's images and ChatGPT

Collecting Data

1 Measure Plant Growth



Use a ruler to measure the height of each plant, and count the number of leaves

2 Attach Leaf to Fluorometer

Clamp a plant leaf into a chlorophyll fluorometer in the clip hole. Shine light on the leaf to measure fluorescence.



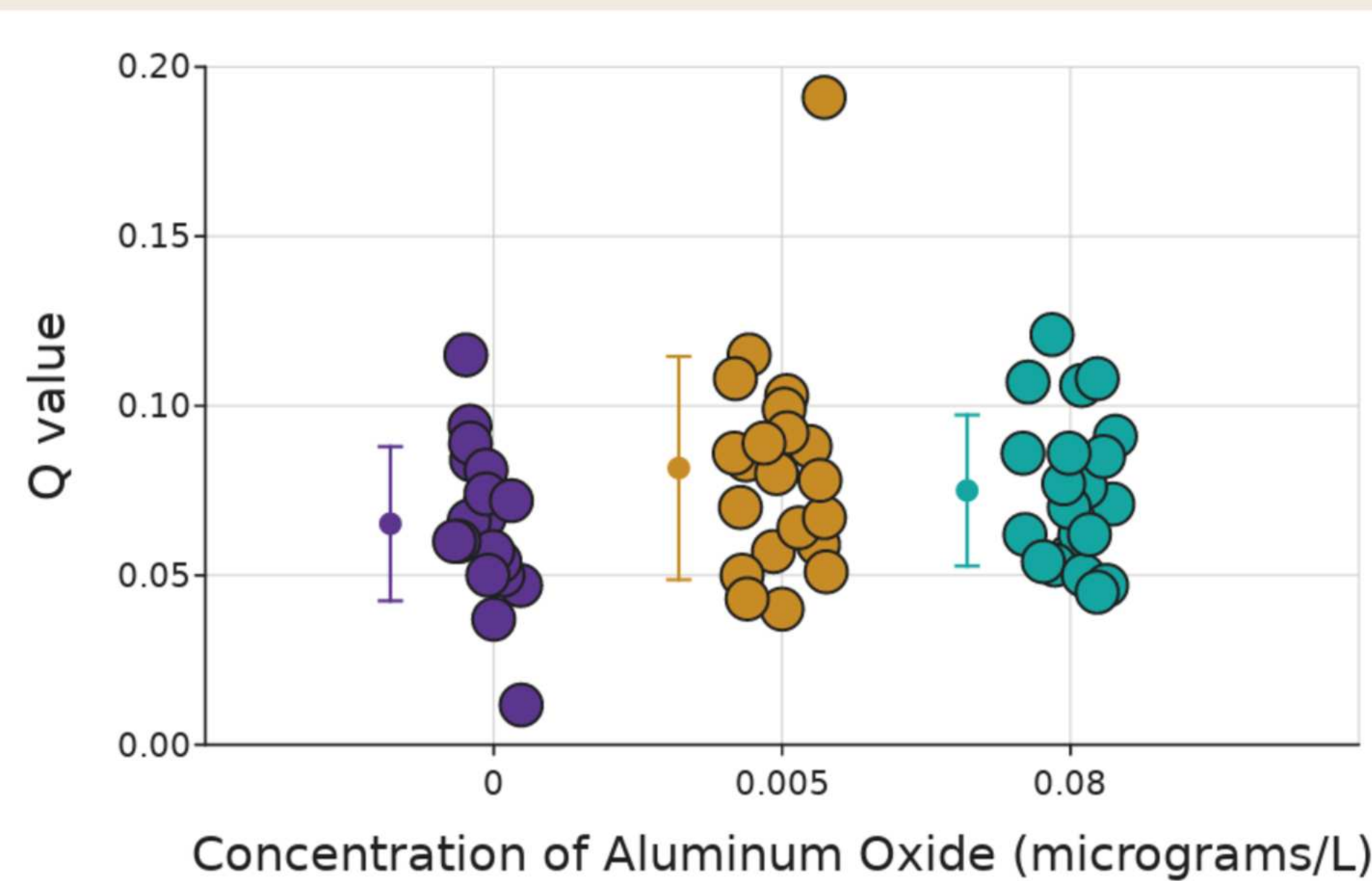
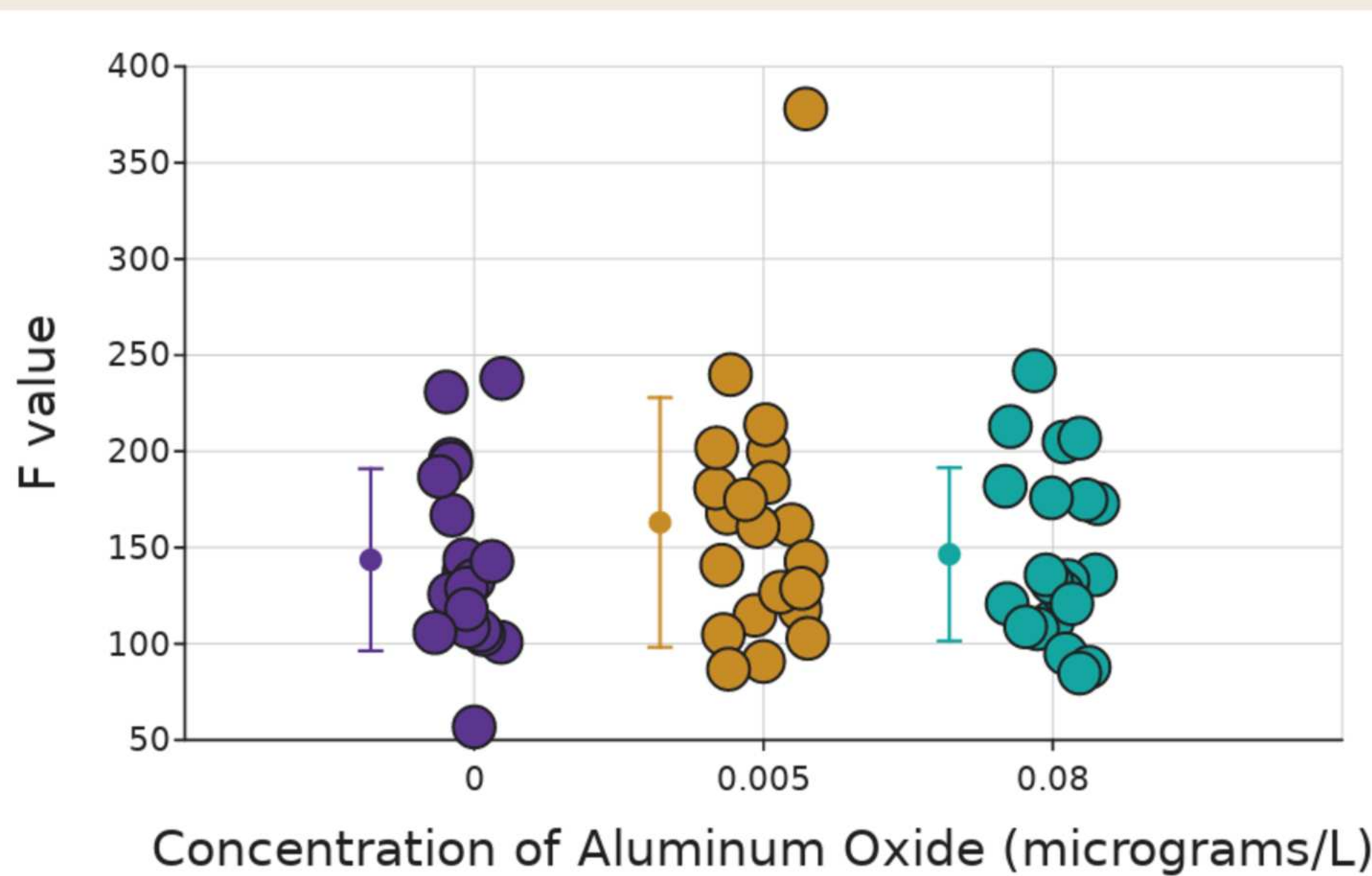
3 Record Fluorometer Data



Hook the fluorometer to a computer, check Quench & F values using software, and record the data.

Graphic created with researcher's images and ChatGPT

Findings



Concentration (micrograms/L)	Height (cm)	Leaves
0	6.34	5.4
0.005	6.78	6.24
0.08	6.92	6.45

Discussion

Statistical Analysis

F values - ANOVA P-Value: 0.46, indicates **little to no difference** between groups

Q values - ANOVA P-Value: 0.16, indicating **some difference that may be present between groups**

Post-hoc test results

0 compared to 0.005: **some indication** that groups might be different

0.005 compared to 0.08: **little to no indication** that the groups might be different

0.08 compared to 0: **no evidence** that the groups may be different

Indications

- Statistical analysis revealed little/no difference in the higher concentration, but a possible difference between the lower concentration and the control
- The result may be due to the **outlier** increasing the average of the F and Q values of the 0.005 concentration
- No strong indications** of the concentration of aluminum oxide impacting the health of photosystem 2
- Growth indicated a pattern between concentrations and height & number of leaves
- Higher concentrations of aluminum oxide resulted in more growth**

Limitations and Future Work

Specific Conditions

- short testing period, only using one species, hydroponic growth conditions rather than natural conditions, and a limited concentration range caused specific conditions for data, leaving the results possibly **unapplicable to other conditions**

Less environmental unpredictability

- Lab-growth conditions do not mimic the natural environment, and **don't mimic natural environmental stressors**

Future work

- longer exposure periods to observe possibly **chronic impacts** of aluminum oxide
- other species of plants to observe if growth patterns are **consistent across species**

References

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